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An adaptive fuzzy handover triggering approach for Long-Term Evolution network

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Abstract: *To cope with the increasing demand for efficient data delivery, self-organizing networks have been introduced in the Long Term Evolution (LTE) system to provide autonomous and flexible mobility management. The existing handover triggering scheme for LTE is not flexible enough to incorporate new performance metrics, and it introduces handover latency. There are studies on non-conventional handoff algorithms for LTE applications, for instance, the fuzzy logic approach. However, the fuzzy logic approach needs regular manual tuning to constantly produce optimal output. In this paper, we address this issue by proposing an adaptive fuzzy logic-based handoff decision algorithm, which can cope with environmental changes and improve efficiency by reducing human intervention. Performance results show that the proposed algorithm can reduce unnecessary handovers by about 20% compared with the fuzzy logic and conventional LTE handover triggering scheme, leading to reduced packet loss rates.*

Keywords: expert system < system, fuzzy system < system, knowledge base < system, artificial intelligence

1. Introduction

Recent studies have shown that the operation of a 3G network, including mobility management, accounts for ~20% of the total operational expenditure (OPEX) cost (Xenakis *et al.*, 2014). This is expected to increase proportionally with the size and sophistication of the technology deployed especially with today's highly complex and advanced cellular networks such as the Long Term Evolution (LTE). One of the major goals in LTE's mobility management is to reduce both the ping-pong handover (PPHO) and the radio link failure (RLF) in order to ensure a manageable OPEX (Akyildiz *et al.*, 2014). The PPHO is an undesirable phenomenon in LTE networks in which a repeated handover occurs between two enhanced Node-Bs (eNBs) because of rapid fluctuations of the received signal strengths from both eNBs at the cell edge. An RLF, on the other hand, occurs when user equipment (UE) is out of the coverage or experiences a severed link for a period (Akyildiz *et al.*, 2014).

The conventional LTE handover decision-making based on the power budget algorithm (PBGT) relies mainly on the reference signal received power (RSRP) (Akyildiz *et al.*, 2014). Handoff triggering is normally regulated by two parameters, namely the time-to-trigger (TTT) and handover margin (HOM). A right balance between the TTT and the HOM is vital to ensure that handovers are not delayed too long while avoiding unnecessary PPHOs. However, optimisation of both parameters requires extensive measuring campaign and statistical analysis (Munoz *et al.*, 2013). There are non-conventional methods to achieve both timely handovers and flexibility, such as the fuzzy logic-based handover algorithm (FLHA) (Zekri *et al.*, 2012). The FLHA also includes the ability to incorporate additional performance metrics for greater choice to configure complex networks. Besides linguistic in nature, which allows for ease of maintenance, the FLHA can make decisions based on multivariate analysis in environments of uncertainty (Zekri *et al.*, 2012). However, the FLHA is not the ubiquitous solution and does not fully mitigate the problems of the PBGT algorithm. It requires human experts and experience to formulate practical fuzzy rules and membership functions through a manual tuning process to obtain the desired output. In addition, the FLHA also does not entirely address the issue of timely triggering of handovers and the flexibility to adapt to rapid changes in the mobile radio channel.

In this work, we develop an extended version of the conventional FLHA algorithm, herein known as the adaptive fuzzy logic handover algorithm (adaptive-FLHA) for LTE networks. Although the FLHA has been proposed to be used in LTE networks, for example, the work by Aziz *et al.* (2010), to our best knowledge, the adaptive-FLHA approach has not been studied for use in LTE systems. In contrast to the FLHA approach, the proposed adaptive-FLHA allows tuning of the fuzzy membership functions

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and rules to be accomplished under minimal human participation, thus potentially reducing OPEX. Furthermore, the proposed algorithm also addresses the drawbacks of the conventional LTE's PBGT algorithm, which relies on tuning of TTT and HOM parameters for optimal performance. While most of the existing studies focus on optimizing the TTT and HOM parameters, for example, the works by Bălan *et al.* (2011) and Munoz *et al.* (2013), the adaptive-FLHA approach aims at reducing handover latency by ensuring timely execution of handovers. Simulation results comparing the performance of the proposed algorithm with that of the FLHA and the conventional LTE's PBGT algorithm in terms of parameters affecting the OPEX will be presented.

The rest of the paper is organized as follows. Section 2 reviews the PBGT algorithm and the FLHA system, while Section 3 presents the proposed adaptive-FLHA algorithm. Section 4 discusses the methodology of measuring the performance, while the performance evaluation results and analysis will be discussed in detail in Section 5. Section 6 concludes the paper with future research direction.

2. Handover decision algorithms

2.1. Long Term Evolution hard handover triggering algorithm

The LTE evolved packet core is first introduced in 2008 in the 3GPP LTE specifications Release 8. It consists of a serving gateway, packet data network gateway and mobility management entity as illustrated in Figure 1. As shown in Figure 2, the LTE handover procedure is divided into three stages: handover preparation, handover execution and handover completion. The signalling protocol for LTE handover initiation phase is based on the concept of a UE-assisted and network-controlled handover. The UE provides performance metric measurements, and the network manages handover execution as illustrated in Figure 2.

During handover preparation, the eNB calculates and makes handover decisions. It also examines the list of candidate eNBs to perform a handover using the LTE's neighbour relationship table (Xenakis *et al.*, 2014). When a handover is required, a request command is sent to the selected eNB, and resource allocation is initiated before proceeding to the handover execution phase. During this phase, the physical connection from the serving eNB is transferred to the targeted eNB via a hard handover mechanism. The handover process concludes with the location update and resource release from the serving eNB.

Table 1 summarizes the notation used in this paper. The existing PBGT algorithm performs handovers to an eNB with a stronger RSRP (Xenakis *et al.*, 2014). In contrast to the received signal strength intensity, the RSRP indicates the actual usable average power within the transceivers' bandwidth by excluding the interference power. To reduce the effect of channel fading and interference, the original received signal strength intensity input will be filtered according to equation 1 to produce the RSRP (Akyildiz *et al.*, 2014),

$$RSRP_n = \left[1 - \left(\frac{1}{2}\right)^{4/K}\right] \times RSRP_{n-1} + \left(\frac{1}{2}\right)^{4/K} \times q_n \quad (1)$$

where $n \in \mathbb{N}$, q_n is the instantaneous value produced at the output of the filter and K is the filter coefficient.

The PBGT algorithm is susceptible to PPHO, which is typically resolved by introducing the HOM and TTT parameters, as shown in Figure 3 (Munoz *et al.*, 2013). The threshold values of the HOM range between 0 and 15 dB, while the TTT have 16 pre-defined values of {0, 40, 64, 80, 100, 128, 160, 256, 320, 480, 512, 640, 1024, 1280, 2560 and 5120} ms. A handover will be triggered when $RSRP_{\text{Target}} > RSRP_{\text{Serving}} + HOM$ for a period equal to or more than TTT (Munoz *et al.*, 2013).

2.2. General architecture of fuzzy logic handover algorithm and handoff factor

Typically, handovers in an FLHA are triggered based on a metric called handoff factor, which is generated from a centroid-based defuzzification process. Handoff factors are a string of numerical values from 0 to 1, where '0' represents

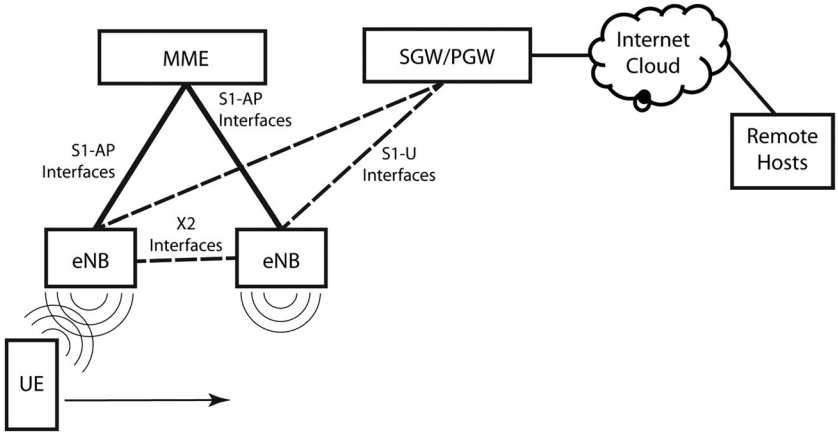


Figure 1: *Long Term Evolution evolved packet core network architecture based on the S1 and X2 interfaces. MME, mobility management entity; eNB, enhanced Node-B; UE, user equipment; SGW, serving gateway; PGW, packet data network gateway.*

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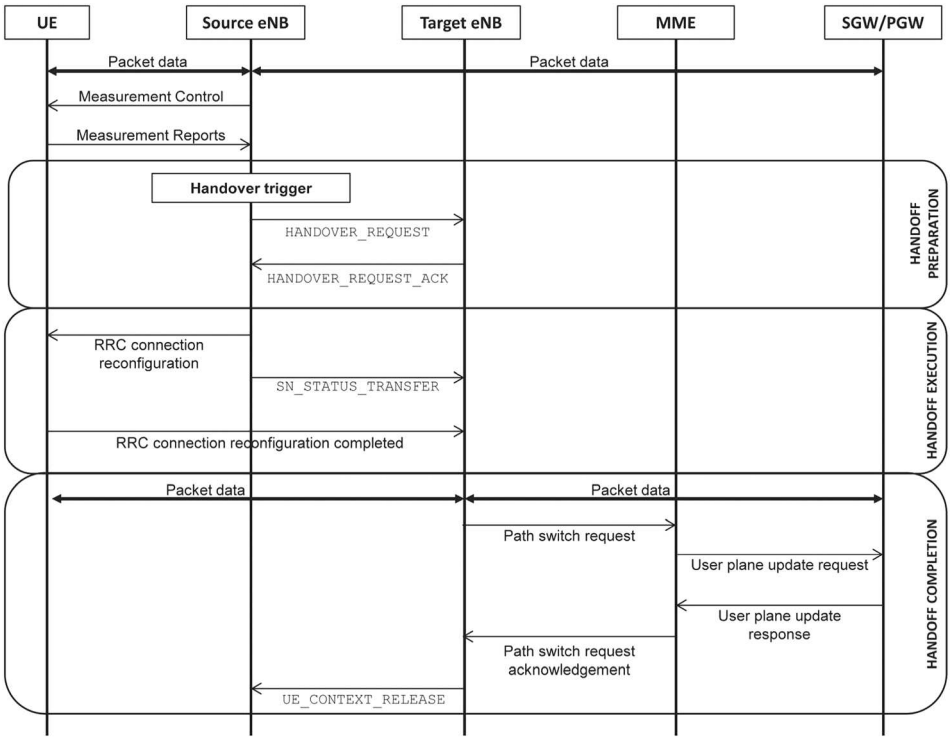


Figure 2: *Signalling for the initiation phase of Long Term Evolution's handover. UE, user equipment; eNB, enhanced Node-B; MME, mobility management entity; SGW, serving gateway; PGW, packet data network gateway; RRC, radio resource control.*

Table 1: *Table of notation*

Symbol	Definition	Symbol	Definition
$RSRP_n$	Output of the RSRP filter	w_i	Firing strength of the adaptive fuzzy logic architecture
q_n	Instantaneous values produced by the RSRP filters	z	Handoff factors
K	Coefficient of the RSRP filters	p	Epoch
x_{RSRP}, x_{BLER}, x_{QoS}	Fuzzy input variables for RSRP, BLER and QoS, respectively	F_p	Training errors for epoch p
p, q, r, s	Consequent parameters	$\Delta\beta$	Training update function
a, b, c	Premise parameters	ζ	Learning rate
M	Membership functions	n	Step size
$\mu_{A1}(x_{RSRP})$	Membership function for RSRP input variable	R	E-model value
$\mu_{B1}(x_{BLER})$	Membership function for BLER input variable	$I_{e,eff}$	Effective equipment factor
$\mu_{C1}(x_{QoS})$	Membership function for QoS input variable	$P_{dejitter}$	Additional buffer packet loss to account for the network jitter

RSRP, reference signal received power; BLER, block error rate; QoS, quality of service.

the least likely to trigger a handover scenario and '1' corresponds to the most likely to trigger a handover scenario. A handoff will be triggered if a handoff factor exceeds the threshold (Pink *et al*., 2012). Figure 4 shows the general architecture of an FLHA.

Without losing generality, we assume that the three membership functions, RSRP, block error rate (BLER) and quality of service (QoS), are part of the fuzzification process. At the input stage, normalized values of the RSRP, BLER and QoS perceived are mapped to their respective degree of the membership function to produce crispy values for processing in the fuzzy inference system. For instance, in Figure 5(a), if the normalized input RSRP is 0.2, it will then be mapped as 0.49 within the domain of 'WEAK', while the normalized RSRP value of 0.9 will produce a crispy value of 0.8. Inside the fuzzy inference system, a series of pre-defined If-Then rules will determine the output values as in Table 2.

3. Proposed Long Term Evolution handover triggering scheme

3.1. Adaptive-fuzzy logic-based handover algorithm using neural networks for handover decision-making

An adaptive-FLHA structure is proposed to execute handover decision-making based on fuzzy logic. In particular,

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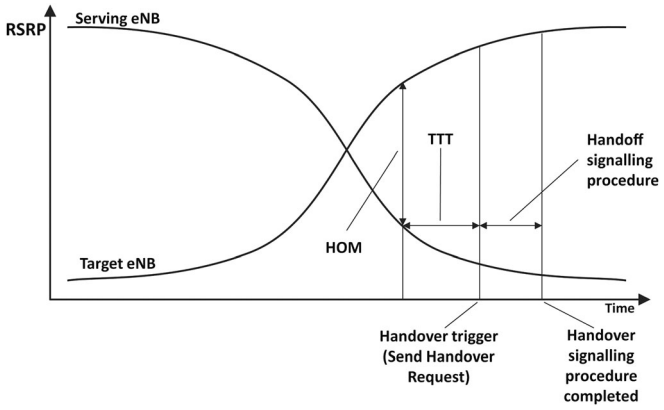


Figure 3: *Hard handover trigger procedure in Long Term Evolution systems. RSRP, reference signal received power; eNB, enhanced Node-B; TTT, time-to-trigger; HOM, handover margin.*

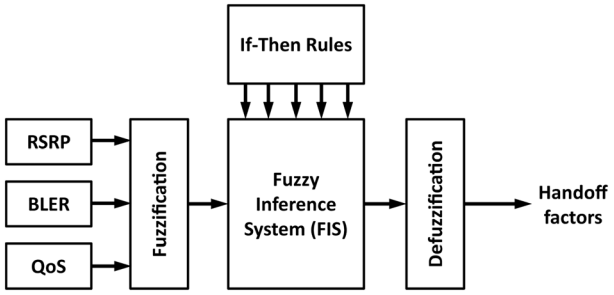


Figure 4: *The block diagram of a fuzzy logic handover decision algorithm. RSRP, reference signal received power; BLER, block error rate; QoS, quality of service.*

the membership functions and rules can be adapted in response to the environmental changes. In contrast to the conventional FLHA, the proposed algorithm enables the rules and membership functions to be adaptive based on the historical data available within the tracking area (TA). This is achieved with the aid of a training algorithm organized into two stages: initialisation and handover trigger. The structure of the adaptive-FLHA is illustrated in Figure 6. The three performance metric inputs considered, x_{RSRP}, x_{BLER} and x_{QoS}, represent the RSRP, BLER and QoS perceived, respectively, and may be further expanded to include more inputs such as channel quality indicator. However, considering too many input variables may desensitize the handover trigger, leading to late handovers that in turn result in RLFs. The BLER is measured at the output of the hybrid automatic repeat request to take advantage of the error correction scheme (Ikuno & Kwan, 2013).

The basic adaptive fuzzy logic structure was first proposed by Jang (1993). This structure uses a training algorithm modified from an artificial neural network to self-optimize the fuzzy inference system. Most of the earlier conventional FLHAs are based on the Mamdani model, which is simpler in design than other models although not computationally efficient. The adaptive-FLHA designed in this paper adopts the Takagi–Sugeno–Kang fuzzy model, which is more computationally efficient (Celikyilmaz & Burhan Turksen, 2008). The proposed If-Then rules of the adaptive fuzzy model are given by

***Rule* 1:** If (x_{RSRP} is A_1) and (x_{BLER} is B_1) and (x_{QoS} is C_1), then $f(i) = p_1\, x_{RSRP} + q_1\, x_{BLER} + r_1\, x_{QoS} + S_1$

⋮

***Rule i*:** If (x_{RSRP} is A_j) and (x_{BLER} is B_j) and (x_{QoS} is C_j), then $f(i) = p_i\, x_{RSRP} + q_i\, x_{BLER} + r_i\, x_{QoS} + S_i$

⋮

Rule U: If (x_{RSRP} is A_3) and (x_{BLER} is B_3) and (x_{QoS} is C_3), then $f(U) = p_U\, x_{RSRP} + q_U\, x_{BLER} + r_U\, x_{QoS} + S_U$

where p_i, q_i, r_i, and s_i are the consequent parameters of the i-th rule and U denotes the number of rules. The consequent parameters are used to adjust the weightage of the rules during the training process. The entire architecture is divided into five layers as illustrated in Figure 6. Layer 1 performs the standard fuzzy logic fuzzification process highlighted as square nodes in Figure 6, in which crispy inputs are converted into non-crispy format using the fuzzy membership functions with linguistic labels. For simplicity, the bell-curved functions shown in Figure 5 are used as the initial membership functions. An example of the mathematical function for a generalized bell function (Soto *et al.*, 2014) based on x_{RSRP} is given as

$$\mu_{A_1}(x_{RSRP}) = \exp\left[-\left(\frac{x_{RSRP} - c_i}{a_i}\right)^{2b_i}\right] \tag{2}$$

where a, b and c are the premise parameters, which are used to adjust the membership functions for optimisation. Most of the earlier work such as Singhrova and Prakash's (2012) adopted the combination of trapezoidal and triangular membership functions (sometimes known as the shouldered membership function). However, this may complicate the adaptive-FLHA training algorithm because it involves multiple layers of training to adjust the parameters of both the trapezoidal and triangular membership functions separately. For this reason, the combination of the generalized bell curves in Figure 5 is used instead because it is the closest approximation to the shouldered functions.

In Layer 2, all signals from Layer 1 are multiplied to every single circle node (labelled Π in Figure 6) to yield w_i the following output:

$$w_1 = \mu_{A_1}(x_{RSRP}) \times \mu_{B_1}(x_{BLER}) \times \mu_{C_1}(x_{QoS}) \tag{3}$$

while Layer 3 nodes (labelled as 'N' in Figure 6) calculate the ratio of the i-th rule's firing strengths in terms of normalized values

$$\overline{w}_i = \frac{w_i}{w_1 + w_2 + \cdots + w_U} \tag{4}$$

before passing the signals to Layer 4, which has a node function given by

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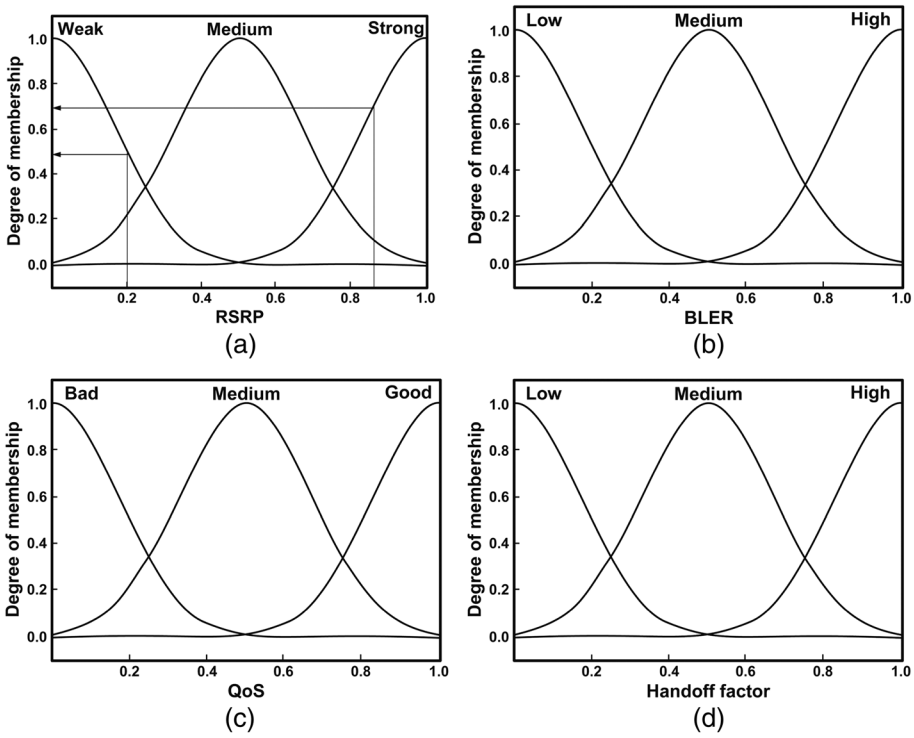


Figure 5: *Membership functions for (a) reference signal received power (RSRP), (b) block error rate (BLER), (c) quality of service (QoS) perceived and (d) the output.*

Table 2: *If-Then rules of fuzzy logic-based handover algorithm*

Rule	If	AND	AND	Then
1	RSRP = high	BLER = low	QoS = good	Handover = low
2	RSRP = high	BLER = low	QoS = medium	Handover = low
⋮	⋮	⋮	⋮	⋮
U	RSRP = low	BLER = high	QoS = bad	Handover = low

RSRP, reference signal received power; BLER, block error rate; QoS, quality of service.

$$\overline{w}_i f(i) = \overline{w}_i (p_i x_{RSRP} + q_i x_{BLER} + r_i x_{Qos} + S_i) \quad (5)$$

Finally, in Layer 5, all signals converge and are summed as a single output, z, expressed as

$$z = \overline{w}_i f(i) = \frac{\sum_i^U \overline{w}_i f(i)}{w_i} \quad (6)$$

where z is the output of the defuzzification process, which is referred as the handoff factor.

3.2. Training algorithm

Training can be categorized into two types: online (pattern learning) and offline training (batch learning). The online training will be triggered when performance is below the acceptable threshold, which is based on historical data, or it can be arbitrarily assigned based on the requirements. This ensures that the membership functions and rules are optimized most of the time. Offline training, on the other hand, will update the parameters when data are available. It uses calibrated models to ensure that handovers are triggered when certain conditions (e.g. terrain or vegetation obstructions) are met, especially when the UEs travel on a fixed route, for example, train routes. However, offline training is not the focus in this paper as it is not an integral part of the proposed algorithm. It can be part of the system's operation and maintenance section, where it can be carried out either autonomously or via system engineers using other form of optimisation techniques such as genetic algorithm.

Both training approaches are accomplished using a hybrid of the steepest decent and the least-square estimator algorithms, enabling fast convergence and a less exhaustive search than the algorithm based on neural networks (Jang, 1993). The output of the hybrid method is given by

$$z = \frac{w_1}{w_1 + \cdots + w_i \cdots + w_U} f(1) + \cdots + \frac{w_i}{w_1 + \cdots + w_i + \cdots + w_U} f(i) + \cdots + \frac{w_i}{w_1 + \cdots + w_i + \cdots + w_U} f(U) \quad (7)$$

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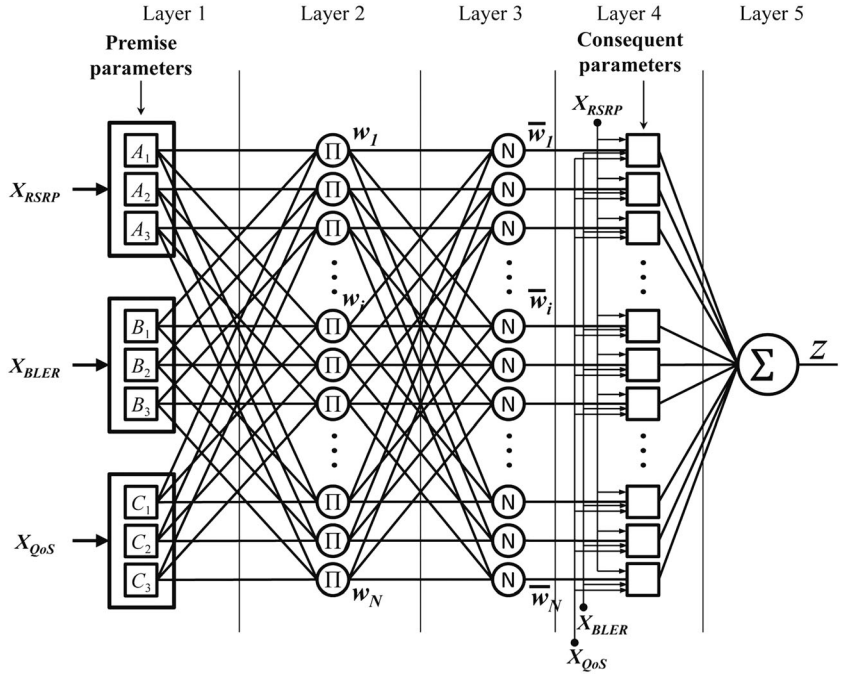


Figure 6: *Architecture of the proposed adaptive fuzzy logic-based handover algorithm.*

The hybrid method employs the forward and backward passes for every round (or epoch, p), and the training update equation is given by

$$\frac{\partial F}{\partial \beta} = \sum_{p=1}^{p} \frac{\partial F_P}{\partial \beta} \tag{8}$$

with the update function $\Delta\beta$ given by

$$\Delta\beta = -\zeta \frac{\partial F_P}{\partial \beta} \tag{9}$$

where ζ is the learning rate and can be expressed as

$$\zeta = \frac{n}{\sqrt{\sum_{\beta} \left(\frac{\partial F}{\partial \beta}\right)^2}} \tag{10}$$

where n is the step size. Following the defuzzification process, the output z in Layer 5 produces the handoff factors.

3.3. Initialisation and handover trigger algorithm

The algorithm for adaptive-FLHA resides in the handover preparation stage and comprises two processes: initialization and handover trigger, as shown in Algorithm I and Algorithm II, respectively. The initialization process is initiated when the UE performs an attach procedure or enters into a new TA (Lim & Hong, 2013). If the UE is new to the TA, online training will commence to adjust the premise and consequent parameters until desired performance is achieved. Otherwise, it will proceed to the periodical UE measurement report (Figure 2).

Algorithm 1: Initialisation stage

```
While (true)
  UE's registration / attach procedure
    if UE is new to TA
      Do adaptive-FLHA training
      Send UE measurement report
    end if
  Send UE measurement report
End
```

Algorithm 2: Handover trigger

```
While (true)
        Send Measurement_Reports
input     : RSRP, BLER, QoS measurements into fuzzier
            If-Then rules consideration
output    : Defuzzifcation = handoff_factor
        if handoff_factor>handoff_threshold then
            eNB check NRT
            Select suitable target eNB
            Send HANDOFF_REQUEST
            Send Path_Switch_Request
        End if
End
```

The handover triggering process is based on the values of handoff factors at the output of the fuzzy defuzzification process as described in equation 6. To control the sensitivity of the handover triggering process, a handover threshold is

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introduced, which can be adjusted automatically or arbitrarily based on the performance requirements in terms of packet loss, PPHO rate, RLFs or call drops, and so on. If the handoff factor exceeds the handover threshold, a handover to the neighbouring eNB with the lowest handoff factor in the neighbour relationship table list will be performed.

4. Handover performance indicators and analysis

4.1. Mean opinion score for VoIP

Conventionally, the mean opinion score (MOS) is used to benchmark the quality of VoIP by assessing the voice quality with a scale between '0' and '5', where '0' denotes the worst quality with '5' being the best. The International Telecommunication Union has devised a tool known as the E-model, which can scientifically calculate the MOS based on the variables that affect the quality of voice received (Jelassi *et al.*, 2012). The E-model is given as follows:

$$R = R_0 - I_s - I_d - I_{e,eff} + A \tag{11}$$

where R_0 is the basic signal-to-noise ratio, I_s represents all the end-to-end impairments related to the voice signal, I_d is the sum of all other impairments that might affect voice quality and A is the advantage factor (Kovac *et al.*, 2011). The parameter $I_{e,eff}$ is called the effective equipment factor and is given by

$$I_{e,eff} = I_e + \left(95 - I_e \cdot \frac{P_{l,eff}}{P_{l,eff} + B_{pl}}\right) \tag{12}$$

where I_e is the impairment factor depending on the VoIP codec used and B_{Pl} is the degree of codec robustness to interference and random losses. The original E-model has a minor drawback because it does not include the effects of network jitter. Although network jitter does not necessarily cause packet losses, it can directly affect the quality of VoIP reception. Based on the work by Kovac *et al.* (2011), we modify the model by including an additional buffer packet loss to account for the network jitter, $P_{dejitter}$, yielding

$$P_{e,eff} = 1 - \left(1 - P_{pl}\right)\left(1 - P_{dejitter}\right) \tag{13}$$

where P_{pl} is the network packet loss.

4.2. Call drop ratio

The call drop ratio (CDR) is useful to quantify the performance of the connection that affects user experience in relation to the RLF and PPHO. The CDR is given as

$$\mathrm{CDR} = \frac{N_{\mathrm{dropped}}}{N_{\mathrm{finished}}} = \frac{N_{\mathrm{dropped}}}{N_{\mathrm{finished}} + N_{\mathrm{success}}} \tag{14}$$

where N_{dropped} is the number of dropped calls as a result of the effect of the RLF and PPHO, N_{success} is the number of successful calls, N_{finished} is the total number of finished calls

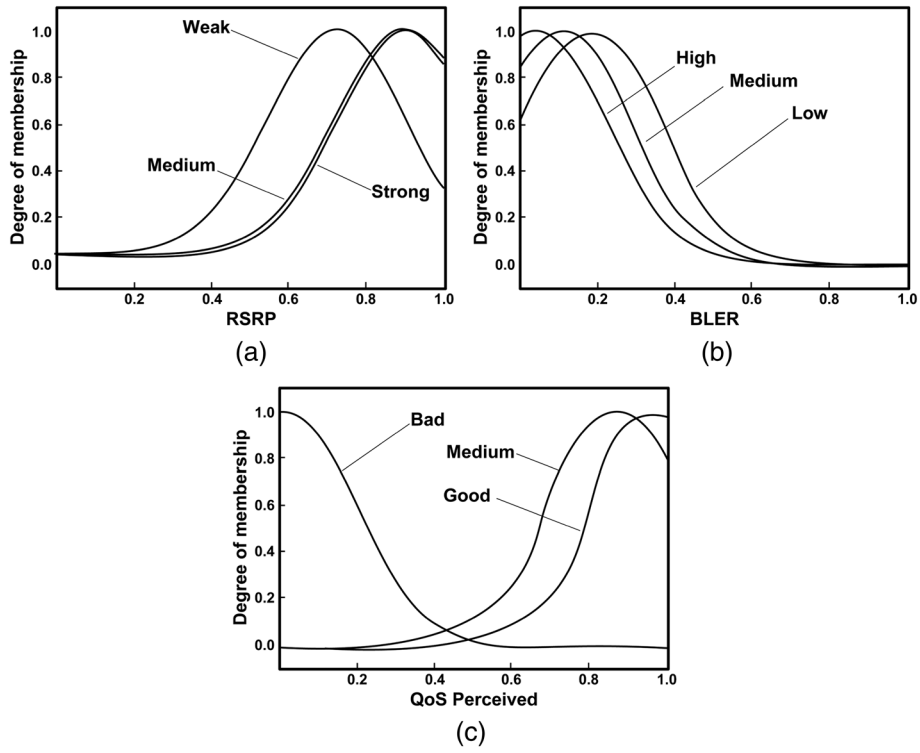


Figure 7: *Membership functions after the training process for (a) reference signal received power (RSRP), (b) block error rate (BLER) and (C) quality of service (QoS) perceived.*

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and N_{handoff} is the number of handovers in the environment (Munoz *et al.*, 2013).

5. Performance evaluation

5.1. Membership functions configuration

In the FLHA, the crisp input of the fuzzy inference system controller is shown in Figure 4, where x_{RSRP} is simulated using a standard path loss model that incorporates Rayleigh fading. The x_{BLER}, on the other hand, is approximated as

$$x_{BLER}(\alpha) = 1 - \sum_{i=0}^{t} \binom{n}{i} p(\alpha)^i [1 - p(\alpha)]^{n-1} \tag{15}$$

where α is the signal-to-noise ratio, $p(\alpha)$ is the coded bit error rate, n is the number of bits per block and t is the number of error bits (Toyserkani *et al.*, 2010). The point of measurement for x_{BLER} is usually carried out by taking BLER measurements after error correction to take advantage of the better BLER performance. The QoS parameter, x_{QoS}, on the other hand, is modelled by the network jitter using the generalized Pareto distribution given as

$$x_{QoS(\eta,\mu,\sigma)}(x) = \frac{1}{\sigma}\left(1 + \frac{\eta(x-\mu)}{\sigma}\right)^{\left(-\frac{1}{\eta}-1\right)} \tag{16}$$

where σ is the standard deviation, η is the shape parameter and μ is the location parameter (Kovac *et al.*, 2011). Each membership function for the metric mentioned earlier uses the untuned hypothetical membership functions shown in Figure 5 for the FLHA. Figure 7 illustrates the trained membership functions after the training process described in Section 3.2; the initial membership functions are shown in Figure 5.

5.2. Scenario configuration

A simulation model has been developed using the Network Simulator 3 (NS-3) platform to evaluate the effectiveness of the proposed adaptive-FLHA. The NS-3 LTE model is based

Table 3: *Simulation parameters*

Parameters	Specifics
Propagation model	Standard path loss model
Operating bandwidth per UE (MHz)	10
Fading model	Rayleigh
Carrier frequency (GHz)	2.6
Number of macro eNB	• 7 (Scenarios A1 and B1) • 3 sectors × 7 cells (for Scenarios A2, B2 and C) • All scenarios are simulated with 2-3-2 cell layout configurations
Distance between eNBs	500 m
UE speed (km/h)	• 60 (Scenarios A and B) • 5–60 (Scenario C only)
Number of UE for Scenarios A1, A2, B1 and B2	Single UE
Number of UEs for Scenario C	{10, 30, 50, 70, 90, 110, 130, 150, 170, 190, 220}
Data (Scenarios A1, A2, B1 and B2)	Infinite (50 kbps)
(Scenario C)	VoIP codec: AMR-WB (12.65 kbps)
Calls arrival (Scenario C only)	Three calls per UE (one call after another)
Calls duration (Scenario C only)	Random 1–30 min per call for each UE, based on exponential distribution
eNB transceiver power (dBm)	46
UE transceiver power (dBm)	2.15
Simulation time(s) (Scenario C only)	300

eNB, enhanced Node-B; UE, user equipment.

Table 4: *Simulation scenarios*

Scenario	Cell configuration	Reference figure	Routes	Mobility model
A1	Single sector with an omnidirectional antenna	Figure 8(a)	eNB-to-eNB	Waypoints {T1, T2, T3, T4, T5}
A2	Three sectors with directional antennas	Figure 8(b)	eNB-to-eNB	Waypoints {S1, S2, S3, S4, S5}
B1	Single sector with an omnidirectional antenna	Figure 8(a)	Moving along the edge of the cell (worst-case scenario)	Waypoints {V1, V2, V3, V4, V5}
B2	Three sectors with directional antennas	Figure 8(b)	Moving along the edge of the cell (worst-case scenario)	Waypoints {R1, R2, R3, R4, R5}
C	Three sectors with directional antennas	—	Random	Random waypoints

eNB, enhanced Node-B.

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on the LTE evolved packet core structure mentioned in Section 2.1. The simulation parameters are presented in Table 3. The simulation compares three different handoff algorithms: the conventional LTE's PBGT handoff algorithm, the FLHA and the proposed adaptive-FLHA triggering scheme. The PBGT algorithm is analyzed based on three different TTT and HOM configurations:

- PBGT-Configuration-A: {TTT = 0 ms, HOM = 0 dB}
- PBGT-Configuration-B: {TTT = 256 ms, HOM = 6 dB}
- PBGT-Configuration-C: {TTT = 5120 ms, HOM = 12 dB}

The performance of these three handover algorithms is compared in the scenarios described in Table 4. Figure 8 shows the radio environment map plot of the scenarios. Figure 8(a) shows seven cells with each eNB designated with an ID from '1' to '7', while Figure 8(b) shows 21 eNBs (7 cells × 3 sectors).

5.3. Handoff factor performance

As described earlier, the product of the defuzzification process is a string of values representing the probability to a handover. This is regulated by introducing a threshold above which a handover will be triggered. In order to compare the handoff factor performance of both the FLHA and the adaptive-FLHA and to obtain the best handoff threshold for both algorithms, an experiment is conducted by moving the UE from one eNB to another. The average packet loss can be calculated from the effect due to delayed handovers, propagation delays and channel conditions. This experiment is repeated with the handoff factor increased from 0.00 to 1.00 with an increment of 0.05. Here, the FLHA achieves the best performance when the handoff factor is 0.70 with a packet loss of 5.93%, while the adaptive-FLHA records a packet loss of 0.45% at a best handoff factor of 0.55. This indicates that the adaptive

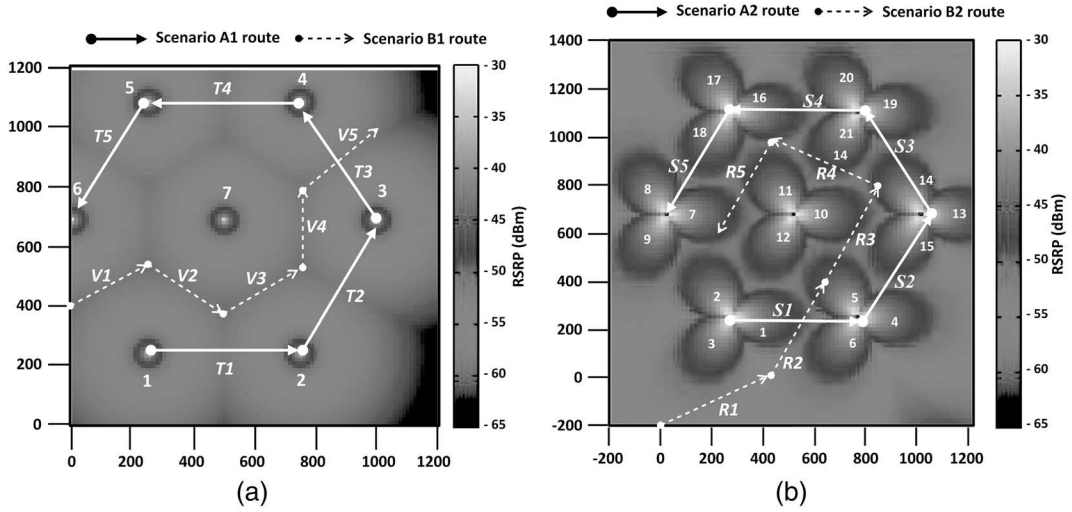


Figure 8: *Radio environment map for (a) Scenarios A1 and B1 and (b) Scenarios A2, B2 and C. RSRP, reference signal received power.*

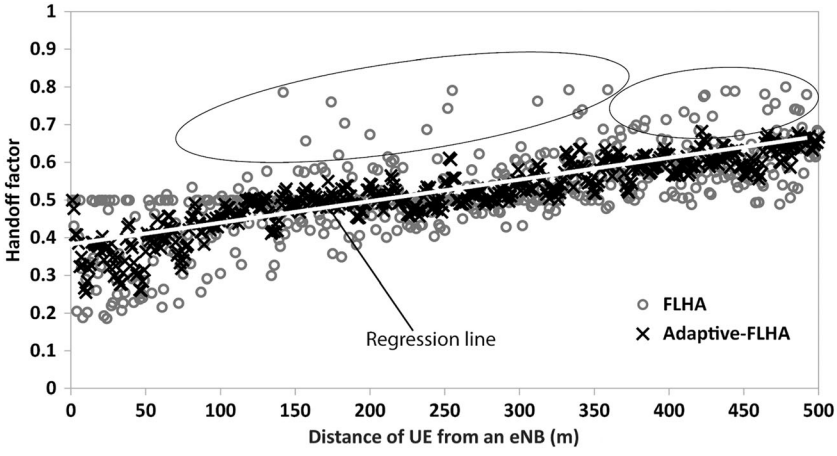


Figure 9: *Comparison of handoff factors for fuzzy logic-based handover algorithm (FLHA) and adaptive-FLHA. UE, user equipment; eNB, enhanced Node-B.*

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element of the adaptive-FLHA has further improved the performance of the FLHA while keeping the total average packet loss below 5% needed for VoIP applications (Jelassi *et al.*, 2012). In the subsequent experiments, the handoff factors of 0.7 and 0.55 are used in the FLHA and the adaptive-FLHA, respectively. This allows both algorithms to be compared fairly using their best handoff factor.

The handoff factor result for both FLHA and adaptive-FLHA is as shown in Figure 9. The performance for adaptive-FLHA is based on the membership function that is presented in Figure 7, which has been tuned in the training process. The results in Figure 9 are obtained by the handoff factors for every 1 m increased between a single eNB and a UE. Naturally, the handoff factors increase with the distance because of the effects of RSRP, which in turn affect the BLER. It is observed that the untuned FLHA causes the handoff factors to spike unnecessarily (as shown inside the oval-shaped lines in Figure 9) and triggers handovers unnecessarily. The regression analysis (indicated as a straight line in Figure 9) also indicates that the *R*-square values of the FLHA and the adaptive-FLHA are 49.64% and 81.71%, respectively, implying higher spread of handoff factors in the FLHA by 32.07%.

5.4. Scenarios A1 and A2: eNB-to-eNB route

The objective of considering Scenarios A1 and A2, with reference to Figure 8(a) and (b), respectively, is to investigate the instantaneous data throughput in ideal situations in which the UEs move from one eNB to another adjacent eNB. The throughput graph is illustrated in Figure 10 for Scenario A1 and in Figure 11 for Scenario A2. In the graphs, minor gaps indicate drastic drops in throughput caused by accumulated delays introduced by the handover latency, which includes delays introduced by TTT. Handovers based on the PBGT-Configuration-A result in small latency, as observed in Figure 10. Close inspection of Figure 10 (labelled 'A') reveals multiple spikes caused by multiple handovers, that is, the PPHOs. The 'one-off' handovers (labelled 'B' in Figure 10) is observed as a single tiny gap at both the 260th and 360th seconds, respectively.

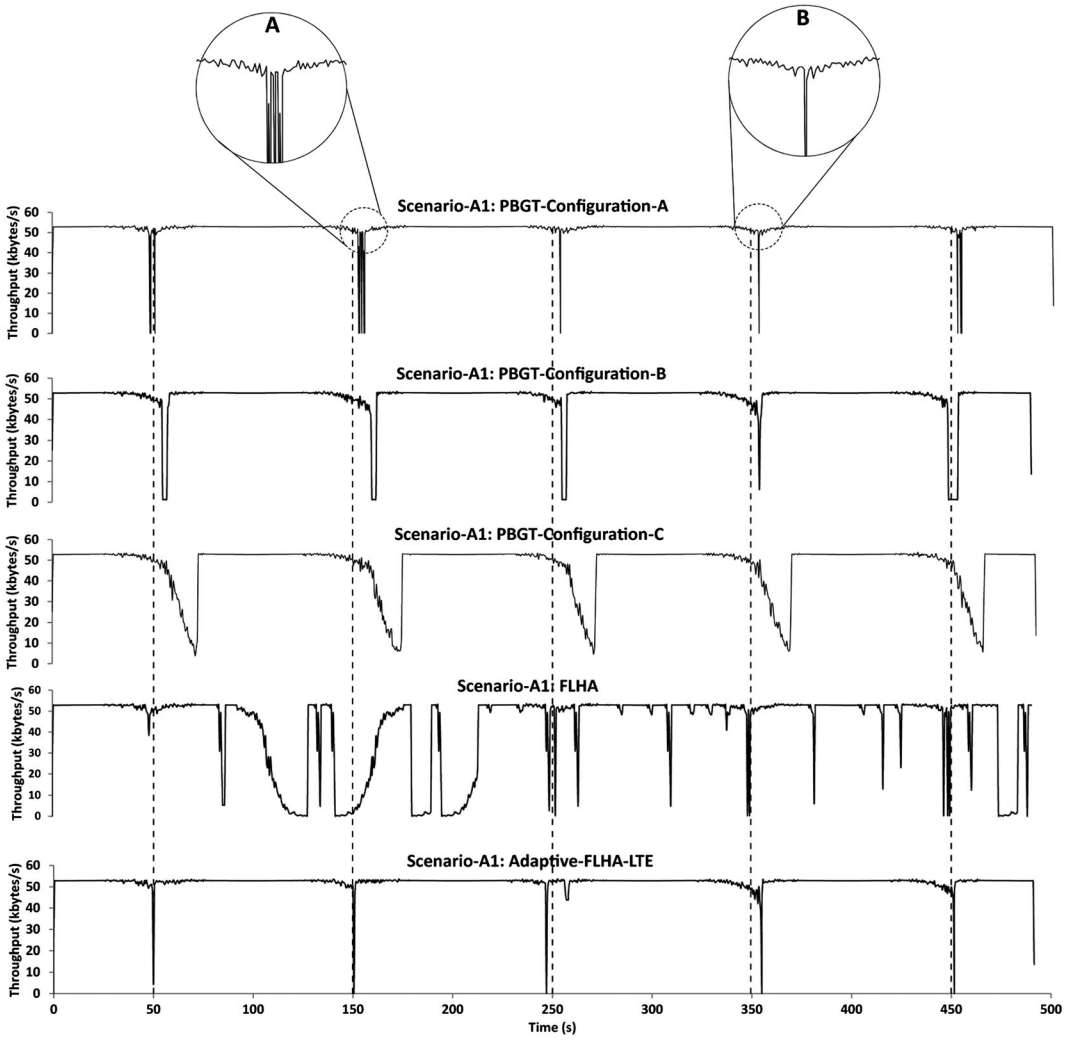


Figure 10: *Throughput performance for Scenario A1. PBGT, power budget algorithm; FLHA, fuzzy logic-based handover algorithm; LTE, Long Term Evolution.*

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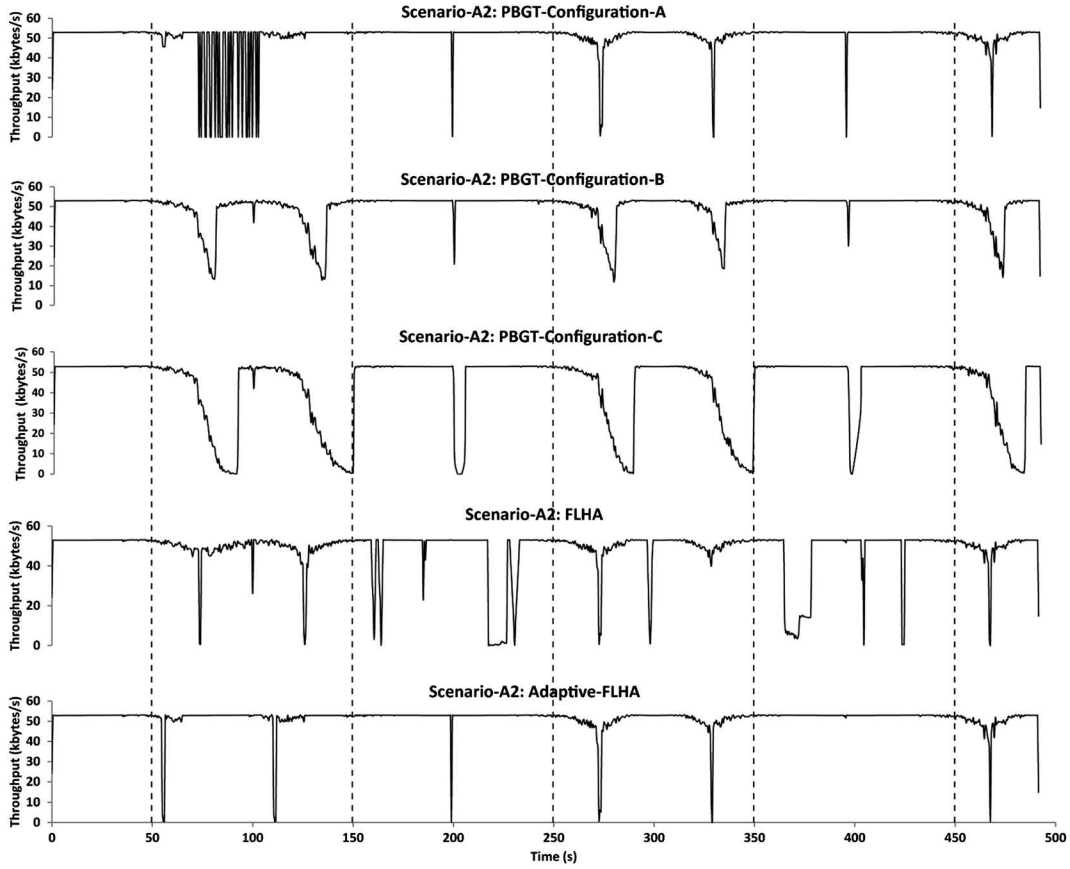


Figure 11: *Throughput performance for Scenario A2. PBGT, power budget algorithm; FLHA, fuzzy logic-based handover algorithm.*

Similar observation can also be seen for the PBGT-Configuration-B and PBGT-Configuration-C. In both Scenarios A1 and A2, PPHOs are eliminated at the expense of handover delays, which lead to potential RLFs. The erratic throughput plot of the FLHA, as depicted in Figure 11, demonstrates that the algorithm behaves unpredictably. For instance, at ~130th and 150th seconds, respectively, handovers are performed unnecessarily. Under normal circumstances, the initial fuzzy logic rules and membership functions defined by human will be fine-tuned and optimized manually. On the other hand, the throughput performance of the proposed adaptive-FLHA shown in Figure 11 indicates that the algorithm adapts well through the training process. In particular, the absence of multiple throughput gaps within a short period indicates that PPHOs have been avoided.

5.5. *Scenarios B1 and B2: worst-case scenario*

Scenarios B1 and B2 account for the worst-case scenarios, as the UEs move along the most undesirable routes, that is, along the cell edge. These routes represent the most vulnerable environments to trigger PPHOs because the RSRP is the lowest along these routes. The throughput performance for Scenario B1 and Scenario B2 is shown in Figures 12 and 13, respectively. In Figure 12, the throughput performance of the PBGT-Configuration-A setting in Scenario B1 shows that PPHOs occur for at least 80% over the simulation period, thus incurring considerable resource wastage.

Similar to Scenarios A1 and A2, Figure 12 shows that the PBGT-Configuration-Band and PBGT-Configuration-C register delayed handovers by 256 and 5120 ms, respectively, which will have severe repercussion on both throughput and packet loss rate performance. The FLHA also behaves unpredictably as handovers are performed unnecessarily, for example, at the 35th and 101th seconds. Also, there are some instances where handovers are not performed when required, for example, the 50th and 230th seconds. This is also expected because both the RSRP and the BLER have dropped below the threshold when the UE is moving along the cell edge, thus making connections unsustainable. Nevertheless, the proposed adaptive-FLHA only suffers from minor PPHOs at the 50th and 120th seconds.

In Scenario B2, it is observed from Figure 13 that PPHOs have been eliminated by both the PBGT-Configuration-B and the PBGT-Configuration-C, with both experiencing late handovers. The FLHA performs handovers to the eNB, which is far away from the UE as observed during the 80th to 101th second, 168th to 167th second and 250th to 260th second, implying that unpredicted handovers have taken place. The adaptive-FLHA, on the other hand, performs 'one-off' handovers over the simulation period.

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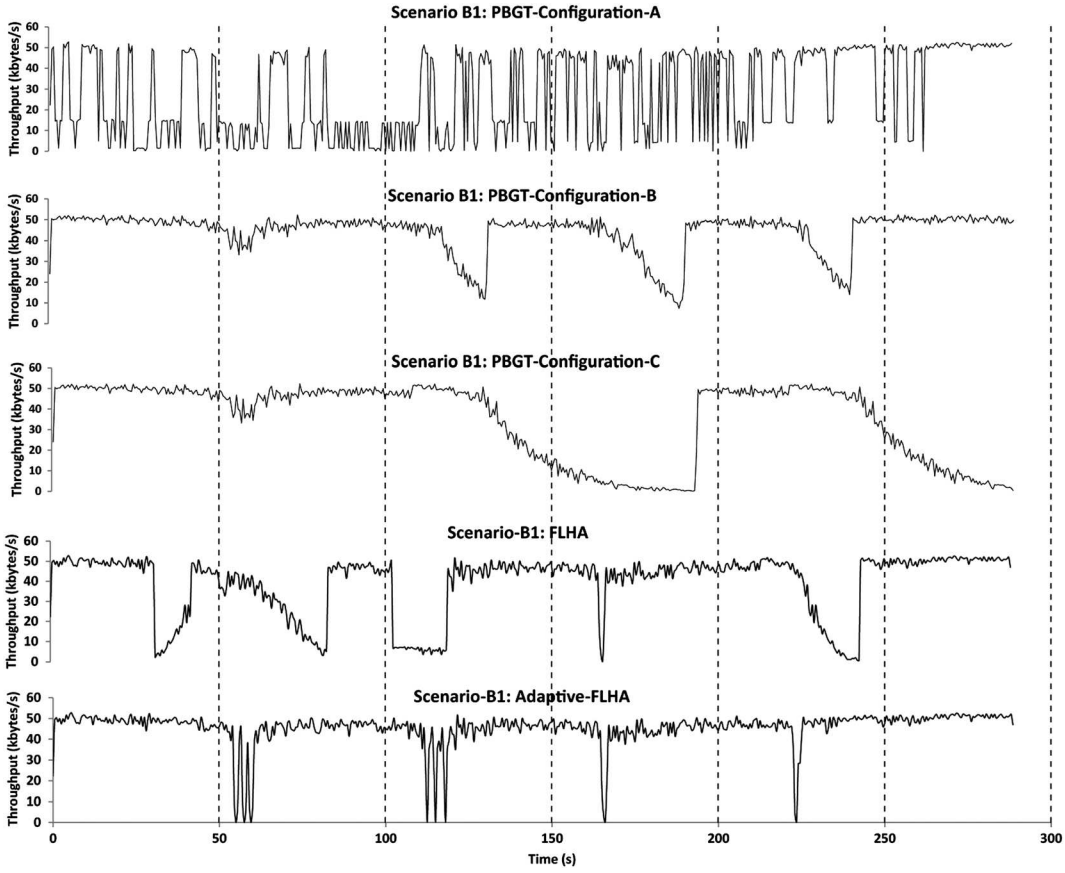


Figure 12: *Throughput performance for Scenario B1. PBGT, power budget algorithm; FLHA, fuzzy logic-based handover algorithm.*

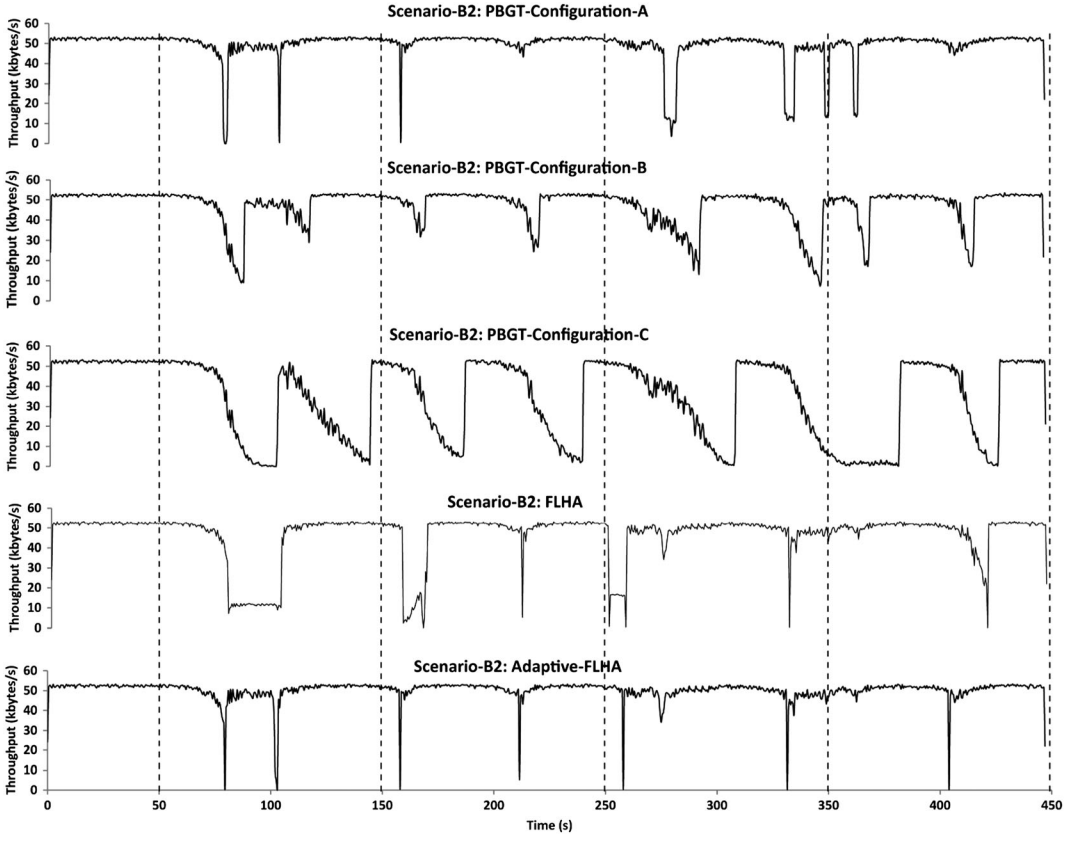


Figure 13: *Throughput performance for Scenario B2. PBGT, power budget algorithm; FLHA, fuzzy logic-based handover algorithm.*

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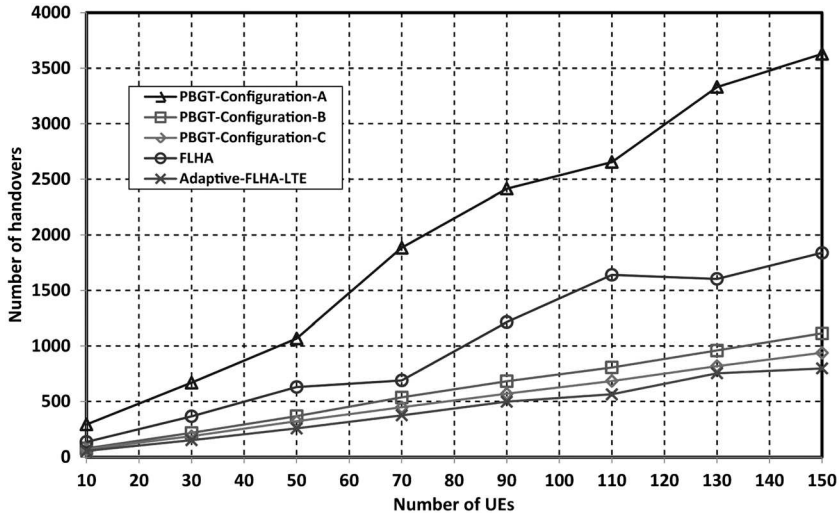


Figure 14: *Average number of handovers per user equipment (UE) versus the number of UEs. PBGT, power budget algorithm; FLHA, fuzzy logic-based handover algorithm; LTE, Long Term Evolution.*

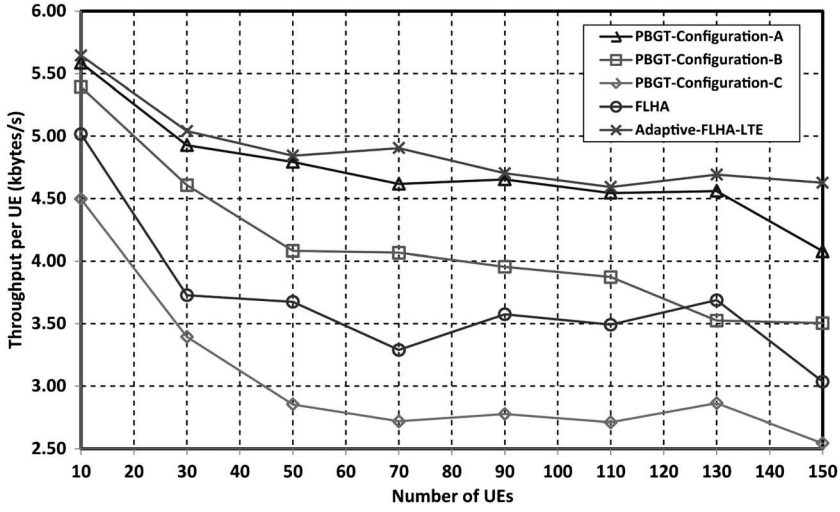


Figure 15: *Throughput performance per user equipment (UE) in Scenario C. PBGT, power budget algorithm; FLHA, fuzzy logic-based handover algorithm; LTE, Long Term Evolution.*

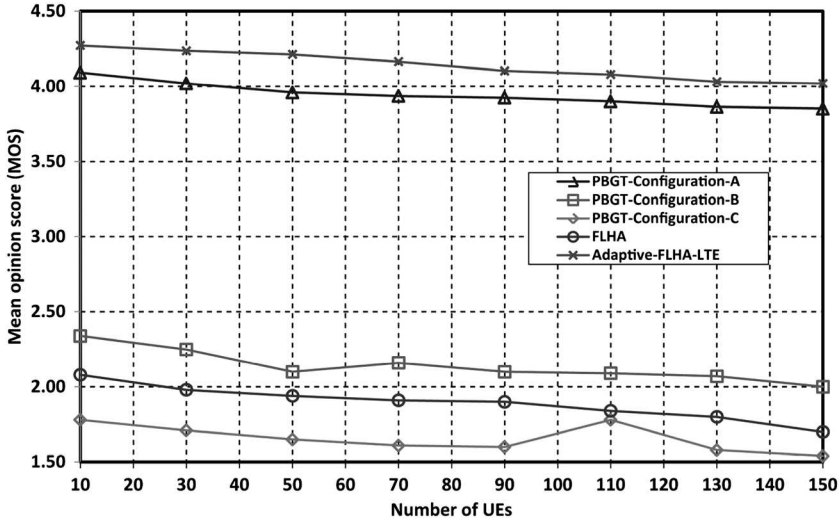


Figure 16: *Average mean opinion score per user equipment (UE). PBGT, power budget algorithm; FLHA, fuzzy logic-based handover algorithm; LTE, Long Term Evolution.*

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5.6. Scenario C: random routes with VoIP traffic

Simulation of Scenario C aims to investigate the overall performance of the algorithms in terms of throughput, packet loss, MOS and CDR. The results are compared with the number of handovers per UE using VoIP traffic. In Figure 14, the adaptive-FLHA records the lowest handover with 798 handovers for 150 UEs (average of 5.34 handovers per UE), while the PBGT-Configuration-A records the highest handover with 3627 handovers for 150 UEs (average of 25.09 handovers per UE). The high handover rate experienced by the PBGT-Configuration-A is attributed to high occurrences of unnecessary handovers. Figure 15 shows that the PBGT-Configuration-A records an average throughput of 4.8114 kb/s, while the adaptive-FLHA achieves an average throughput of 4.8807 kb/s. The 0.0693-kb/s difference in throughput between the two algorithms is due to the packet loss during the link-switching delay caused by the handover process. The reduction of handovers by 80% while maintaining throughput performance demonstrates the effectiveness of the adaptive-FLHA. In effect, this contributes to a lower OPEX because unnecessary resource wastage in handover signalling has been avoided.

At another extreme, the PBGT-Configuration-C inevitably delays handovers even when truly needed, because a trigger will only be executed when the delay TTT and the margin HOM are more than 5120 ms and 12 dB, respectively. In effect, it reduces the handovers per UE to 6.39 compared with 25.09 per UE in the PBGT-Configuration-A. As a result,

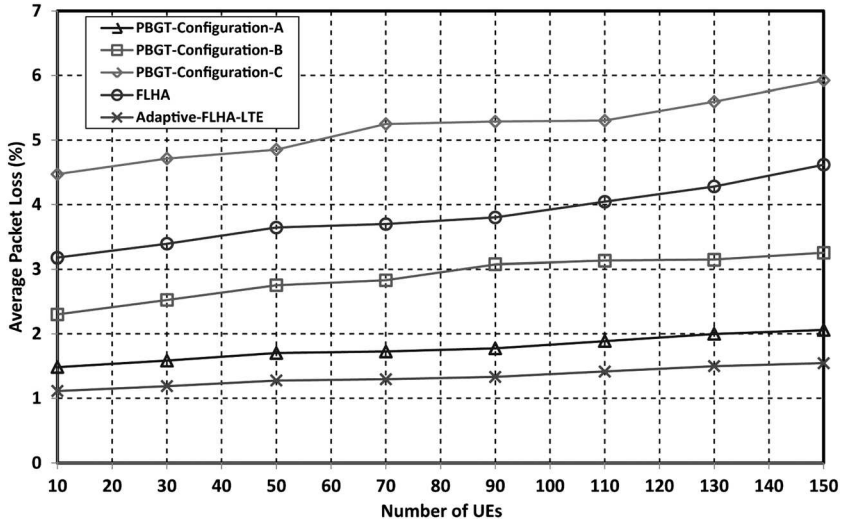


Figure 17: *Average packet loss per user equipment (UE). PBGT, power budget algorithm; FLHA, fuzzy logic-based handover algorithm; LTE, Long Term Evolution.*

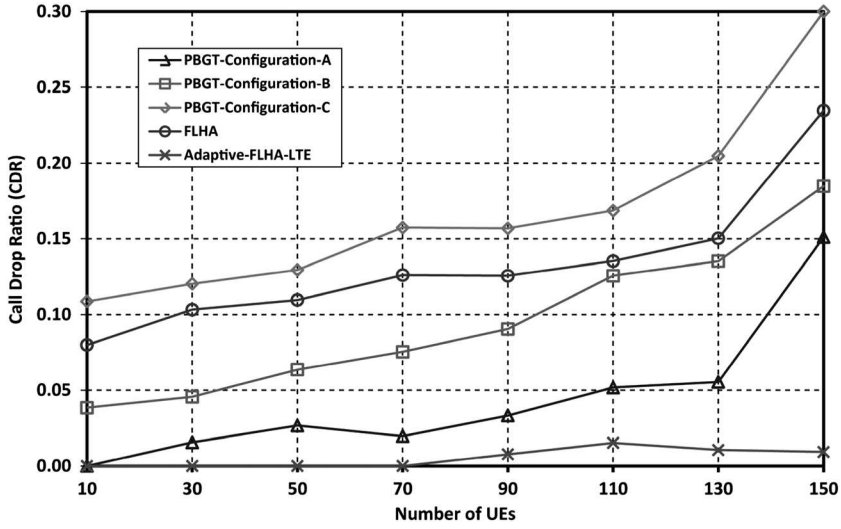


Figure 18: *Call drop ratio per user equipment (UE). PBGT, power budget algorithm; FLHA, fuzzy logic-based handover algorithm; LTE, Long Term Evolution.*

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packets have been lost because of the effect from low RSRP because UEs are moving away from the servicing eNB.

Despite the fact that the packet loss rate has a direct effect on the MOS, the results in Figures 16 and 17 are not linearly correlated. This is because the MOS is also affected by various QoS parameters such as jitter, network delay and bandwidth (Jelassi *et al*., 2012). The proposed adaptive-FLHA achieves a MOS value of 3.5 as shown in Figure 14, while the PBGT-Configuration-A results in a slightly lower MOS value. This shows that the average MOS of the proposed adaptive-FLHA is acceptable for VoIP applications. Other TTT and HOM settings record MOS values below 2.5, which may result in service dissatisfaction (Anehill *et al*., 2012) due to the higher packet loss rate as shown in Figure 17.

It is observed in Figure 17 that the PBGT-Configuration-A records a higher average packet loss rate of 1.74% compared with 1.30% for the adaptive-FLHA due to unnecessary handovers. Although the difference is only 0.44%, it is noteworthy that a reduction of packet loss by 0.1% will significantly degrade the QoS of LTE's VoIP applications (Singh *et al*., 2014). The CDR analysis also reveals that the number of dropped calls due to RLFs and PPHOs is reduced using the proposed adaptive-FLHA scheme, as shown in Figure 22. The PBGT-Configuration-C records the highest dropped calls because of handover delays, which cause unsustainable connections, as similarly observed in Figures 12(c) and 18(c).

6. Conclusion

The existing LTE's PBGT handover algorithm presents two major problems, namely untimely handover and lack of flexibility to adapt to environmental changes. The FLHA can be deployed to mitigate these problems to some extent. However, the FLHA requires manual tuning of the fuzzy membership functions and rules continuously to keep up with the dynamics of the mobile communications. To ensure the fuzzy logic parameters are regularly updated to the latest environment, the current work proposes an adaptive-FLHA that has the capability to improve performance by enabling autonomous tuning with minimal human participation. This is made possible with a well-coordinated training algorithm in place within the adaptive-FLHA to execute training when necessary. Performance evaluation results demonstrate an improvement in the proposed adaptive-FLHA over other handover algorithms particularly in terms of accuracy of handover decisions, thus reducing the handover latency. Deployment of the adaptive-FLHA also results in a reduction of handovers by at least 20% while maintaining the throughput, thus effectively reducing the OPEX. Although the study presented in this paper is new to LTE systems, it is specifically designed for homogeneous applications and can be expanded further to LTE's heterogeneous networks. This requires different strategies and more complex handover procedures to cater to the diverse requirements of various technologies.

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